

While anidados: enunciados y soluciones

Solución = salida impresa por el programa.

1)

```
a = 0
x = 3
y = 4
i = 0
while i < x:
    j = 0
    while j < y:
        a = a + 1
        j = j + 1
    i = i + 1
print(a)
```

Salida: 12

2)

```
s = 0
n = 4
i = 1
while i <= n:
    j = 1
    while j <= i:
        s = s + j
        j = j + 1
    i = i + 1
print(s)
```

Salida: 20

3)

```
x = 17
y = 5
q = 0
while x >= y:
    j = 0
    while j < y:
        x = x - 1
        j = j + 1
    q = q + 1
print(x, q)
```

Salida: 2 3

4)

```
c = 0
n = 5
i = 0
while i < n:
    j = 0
```

```
        while j <= i:
            c = c + ((i + j) % 2 == 0)
            j = j + 1
        i = i + 1
    print(c)
```

Salida: 9

5)

```
ok = False
n = 45231
i = 0
while n > 0 and not ok:
    d = n % 10
    m = n // 10
    j = 0
    while j < 1 and m > 0:
        e = m % 10
        ok = (10 * d + e) == 23 or ok
        m = m // 10
        j = j + 1
    n = n // 10
print(ok)
```

Salida: False

6)

```
b = 0
m = 3
n = 3
i = 0
while i < m:
    j = 0
    while j < n:
        if i == 0 or j == 0 or i == m - 1 or j == n - 1:
            b = b + 1
        j = j + 1
    i = i + 1
print(b)
```

Salida: 8

7)

```
acc = 0
a = 7
b = 9
while b > 0:
    if b % 2 == 1:
        j = 0
        while j < a:
            acc = acc + 1
            j = j + 1
        k = 0
        t = 0
        while k < a:
            t = t + 1
            k = k + 1
        a = t + a
        b = b // 2
print(acc)
```

Salida: 63

8)

```
x = 64241
c = 0
while x > 0:
    d = x % 10
    y = x // 10
    ok = True
    while y > 0:
        ok = ok and (y % 10) < d
        y = y // 10
    c = c + ok
    x = x // 10
print(c)
```

Salida: 1

9)

```
b = 0
h = 3
i = 1
while i <= h:
    j = 0
    while j < 2 * i - 1:
        b = b + 1
        j = j + 1
    i = i + 1
print(b)
```

Salida: 9

10)

```
s = 0
h = 3
i = 1
while i <= 2 * h - 1:
    a = i
    if i > h:
        a = 2 * h - i
    j = 1
    while j <= a:
        if (i + j) % 2 == 0:
            s = s + 1
        else:
            s = s - 1
        j = j + 1
    i = i + 1
print(s)
```

Salida: 3

11)

```
i = 2
n = 30
ok = False
while i <= n and not ok:
    s = 0
    j = 1
    while j < i:
        if i % j == 0:
            s = s + j
        j = j + 1
    ok = s == i or ok
    i = i + 1
print(ok)
```

Salida: True

12)

```
L = 3
x = 0
y = 0
```

```
dx = 1
dy = 1
t = 0
while t < 10:
    i = 0
    while i < L:
        x = x + dx
        if x == L or x == -L:
            dx = -dx
        i = i + 1
    j = 0
    while j < L:
        y = y + dy
        if y == L or y == -L:
            dy = -dy
        j = j + 1
    t = t + 1
print(x == y)
```

Salida: True

13)

```
s = 0
N = 4
i = 1
while i <= N:
    j = 1
    while j <= N - i + 1:
        if (i * j) % 3 == 0:
            s = s + i
        else:
            s = s + j
        j = j + 1
    i = i + 1
print(s)
```

Salida: 20

14)

```
r = 1
b = 3
e = 3
i = 0
while i < e:
    j = 0
    t = 0
    while j < b:
        t = t + r
        j = j + 1
    r = t
    i = i + 1
print(r)
```

Salida: 27

15)

```
p = 0
n = 3
i = 1
while i <= n:
    j = 1
    while j <= n:
        h = (i * j) % 5
        ok = True
        if i > 1:
            ok = ok and (((i - 1) * j) % 5) < h
        if j > 1:
            ok = ok and ((i * (j - 1)) % 5) < h
        if ok:
            p = p + 1
        j = j + 1
    i = i + 1
print(p)
```

Salida: 7